Data Analysis Plan

Introduction: The Framingham Heart Study is a long-term prospective cohort study initiated in 1948 to investigate the etiology of cardiovascular disease in a community population in Framingham, Massachusetts. The subset of data used here includes 4,434 participants examined across three clinic visits approximately 6 years apart (roughly 1956–1968), with up to 24 years of follow-up for cardiovascular outcomes including stroke, myocardial infarction, angina, and death. The purpose of this report is to develop the analysis plan and present preliminary descriptive results to address the hypotheses: (1) Baseline risk factors including age, diabetes status, systolic blood pressure, and potentially prevalent coronary heart disease, antihypertensive medication use, smoking status, total cholesterol, and BMI are independently associated with time to incident stroke within 10 years of baseline, and these associations differ between males and females. (2) Systolic blood pressure and diabetes status change meaningfully across the three examination periods, suggesting that a time-varying covariate approach may provide a more accurate characterization of stroke risk than a baseline-only model in future analyses.

Methods

Data Cleaning: The analytic cohort was restricted to Period 1 (baseline) observations among participants free of prior stroke at baseline, yielding an initial sample of 4,434 participants. Candidate covariates include age, systolic blood pressure, BMI, total cholesterol, diabetes status, current smoking status, antihypertensive medication use, and prevalent coronary heart disease. A complete case analysis was performed, yeilding a final analytic sample of 4,275 participants (1,897 males, 2,378 females). Follow-up time was measured in days

from baseline and capped at $365.25 \times 10 = 3{,}652.5$ days, using 365.25 days per year to account for leap years, to restrict the analysis to 10 years. Participants who experienced a non-stroke event (including death) or remained event-free beyond 10 years were treated as censored. Biologically implausible values will be reviewed and excluded.

Data Analysis: Cox proportional hazards regression will be used to model time to incident stroke using baseline covariates only. Competing risks (e.g., death, myocardial infarction) will not be formally modeled and all non-stroke events will be treated as censored. This will be noted as a limitation. The correlation between systolic blood pressure and antihypertensive medication use will be assessed to evaluate portential collinearity and inform final covariate selection. All analyses will be stratified by sex, with separate models fit for males and females. The final covariate sets may differ between sexes as the goal is to identify risk factors most predictive of stroke within each sex. The proportional hazards assumption will be evaluated using Schoenfeld residuals. Based on the final models, 10-year stroke probabilities will be estimated for clinically meaningful risk profiles. To inform future analyses, systolic blood pressure and diabetes prevalence will be summarized descriptively across examination periods by sex to assess potential time-varying effects.

Preliminary Descriptive Analysis: Baseline characteristics were summarized in Table 1 stratified by sex. with continuous variables reported as mean (SD) and median [min, max] and categorical variables as count (%). Exploratory figures include a boxplot of systolic blood pressure by antihypertensive medication use (Figure 1), histograms of systolic blood pressure by sex (Figure 2), and a scatterplot of age versus systolic blood pressure with a loess smoother (Figure 3). All analyses were conducted using R.

Table 1: Baseline Characteristics by Sex

	Male	Female	Overall
	(N=1897)	(N=2378)	(N=4275)
Age (years)			
Mean (SD)	49.7 (8.71)	49.9 (8.59)	49.8 (8.64)
Median [Min, Max]	49.0 [33.0, 69.0]	49.0 [32.0, 70.0]	49.0 [32.0, 70.0]
Systolic BP (mmHg)			
Mean (SD)	132 (19.3)	134 (24.3)	133 (22.2)
Median [Min, Max]	129 [83.5, 232]	129 [83.5, 295]	129 [83.5, 295]
BMI (kg/m²)			
Mean (SD)	26.2 (3.40)	25.6 (4.52)	25.8 (4.07)
Median [Min, Max]	26.1 [15.5, 40.4]	24.8 [16.0, 56.8]	25.5 [15.5, 56.8]
Total Cholesterol (mg/dL)			
Mean (SD)	234 (42.5)	240 (46.1)	237 (44.6)
Median [Min, Max]	231 [113, 696]	237 [129, 600]	234 [113, 696]
Coronary Heart Disease			
No	1784 (94.0%)	2314 (97.3%)	4098 (95.9%)
Yes	113 (6.0%)	64 (2.7%)	177 (4.1%)
Diabetes			
No	1841 (97.0%)	2321 (97.6%)	4162 (97.4%)
Yes	56 (3.0%)	57 (2.4%)	113 (2.6%)
Current Smoker			
No	750 (39.5%)	1420 (59.7%)	2170 (50.8%)
Yes	1147 (60.5%)	958 (40.3%)	2105 (49.2%)
Antihypertensive Medication			
No	1857 (97.9%)	2286 (96.1%)	4143 (96.9%)
Yes	40 (2.1%)	92 (3.9%)	132 (3.1%)
Incident Stroke within 10 Years			
No	1849 (97.5%)	2321 (97.6%)	4170 (97.5%)
Yes	48 (2.5%)	57 (2.4%)	105 (2.5%)

Figure 1. Baseline Systolic Blood Pressure by Antihypertensive Medication

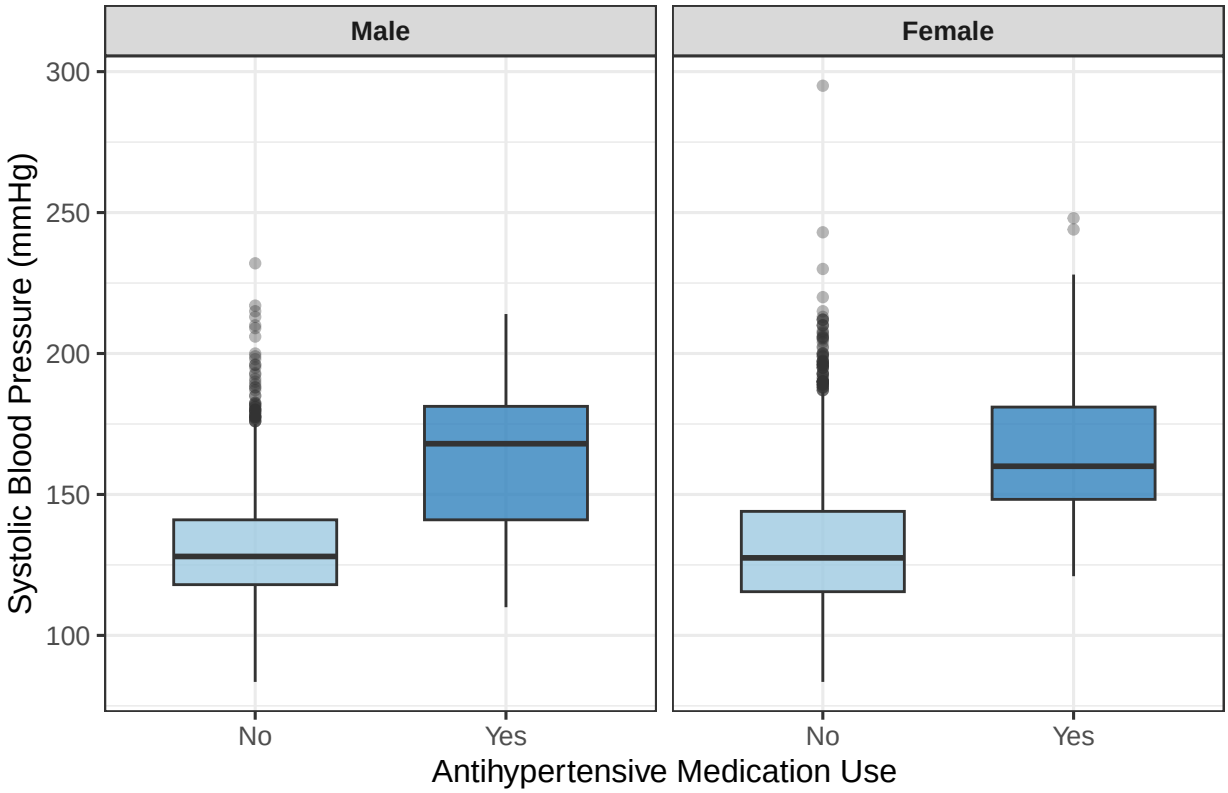

Figure 2. Distribution of Baseline Systolic Blood Pressure by Sex

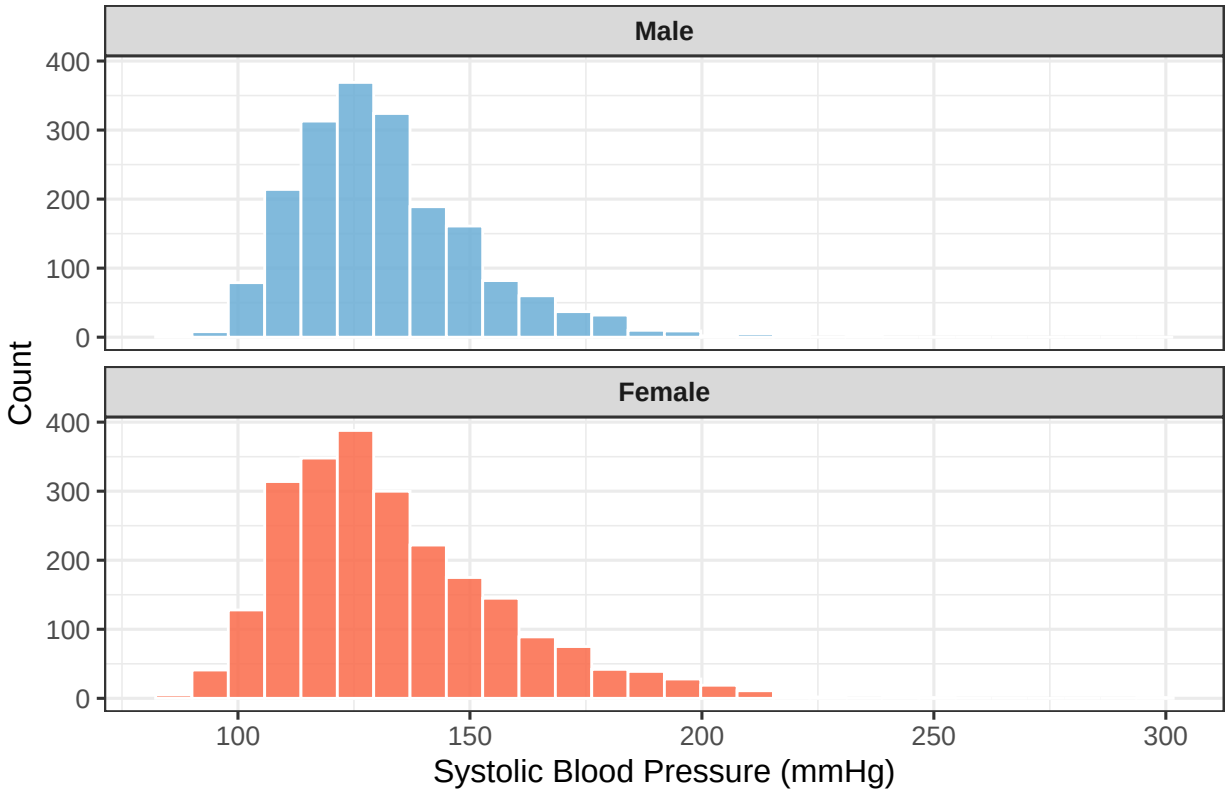
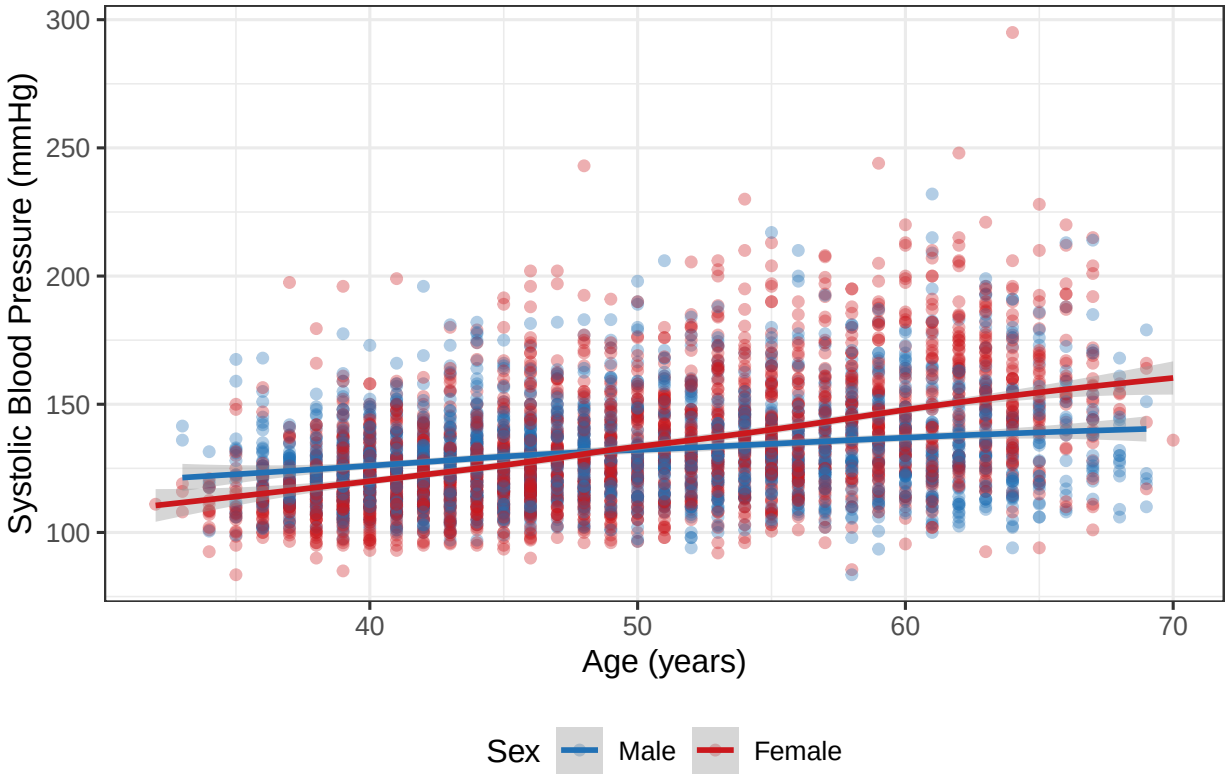
Figure 3. Relationship Between Age and Baseline Systolic Blood Pressu